threading, in synthesis; when we think about using multiple CPUs across several machines communicating through a network, then we are instead referring to what's called *distributed computing*. Figure 13.1 shows a diagram that illustrates this difference.

Parallel computing is limited by the number of CPUs on a single machine, while distributed computing is a better approach to scale out systems and process huge datasets; on the other hand, if a dataset can fit into a single machine's memory, the parallel computing approach results sensibly faster, since the processes can communicate through shared memory, while nodes in distributed systems need to exchange information through a network (at the time of writing, the latency[2] is 100ns vs 150ms[3], so a factor 10^6)

Here in this chapter, we'll often use the term parallel computer to refer to both: the computational models we present are software abstractions that could run seamlessly on threads on a single machine or on a distributed system, and the only discriminant would be the size of the input and the resources needed, not the algorithms we use.